Week 6 Methodology: Supply-Demand Gap Analysis

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Bottom line. The engine joins online vacancy postings and published graduate counts on Week 4 taxonomy domains and reports, for each domain, the gap between its share of demand and its share of supply. On the current sample (13 postings, 3,585 graduates) the largest positive gap is DOM-10 Transport, logistics and supply chains (+10.2 pp) and the largest negative gap is DOM-07 Public policy and governance (-16.0 pp). Only DOM-11 (0 postings, 20 graduates) has an interval excluding zero, and it is LOW SUPPORT, so no category gap is established. The sample's STEM-to-humanities ratio is 0.11:5 against NDPIV's 2:5 current and 3:5 target; this is not comparable, because the supply sample is two management/law-heavy institutions. The value of Week 6 is a working, traceable engine that will produce defensible numbers once Week 3 collection reaches the charter's 2,000 postings.

1. Versions that produced this result

Input	Version / file	SHA-256 (first 16)
Demand + supply records (Week 3)	data/processed/week5_working_sample.csv (Week 3 pipeline_version 0.1.0, working split (week5_heldout.md))	2eacce913588b697
Taxonomy (Week 4)	WEEK 4 - SKILLS AND OCCUPATION TAXONOMY.zip::WEEK 4 SKILLS AND OCCUPATION TAXONOMY.xlsx, v0.1, provisional	019d86d5d85e813f
Extraction output (Week 5)	data/processed/week5_extractions_v1.csv, v1.0-gazetteer-spacy	7d43dff685aa68fc
Mapping rules (Week 6)	config/week6/mapping_rules.csv	41d2fc88362d25e9
Category attributes	config/week6/categories.csv	60f9620fb1481768
Engine config	config/week6/gap_engine.yaml	4115d1eb94487371

The Week 8 held-out split (17 records) is excluded by design and the engine refuses to read it, so Week 8 extraction evaluation stays clean. run_manifest.json stores every hash above plus the hash of every output.

2. Units, population, geography and period

	Demand	Supply
Unit	One de-duplicated online or district vacancy posting	One graduate on a published graduation list
Population	Postings captured from BrighterMonday (12) and Lyantonde District (3) for jobs in Uganda	Graduates of UMI 22nd ceremonies and IUIU 34th ceremony (Kampala campus)
Geography	Uganda; 2 postings located outside Uganda are out of scope	Kampala-based campuses of 2 institutions
Period	Posted 7-21 Oct 2026 (estimated from relative dates); retrieved 21 Oct 2026	Ceremonies Mar-Apr 2025 (UMI) and 20 Dec 2025 (IUIU)
Numerator	Postings mapped to category c (fractional)	Graduates mapped to category c (fractional)
Denominator	All mapped in-scope postings: 13	All mapped graduates: 3,585
Aggregation	Sum of weights by Week 4 domain	Sum of graduate_count x weight by Week 4 domain

Postings are counted once each, not by number of positions, because BrighterMonday does not publish a vacancy count (12 of 15 missing). The demand and supply periods differ by about a year; graduates of 2025 are treated as the supply entering the 2026 market.

3. Primary gap formula and why

For each category c: $\mathbf{D_c}$ = demand units in c / all mapped demand units; $\mathbf{S_c}$ = graduates in c / all mapped graduates; $\mathbf{G_c = 100 \times (D_c - S_c)}$ percentage points. A positive gap means the category is a larger share of the demand sample than of the graduate sample (possible under-supply relative to this demand); a negative gap means the reverse.

Why a share difference: postings and graduates are different units with very different totals, so raw counts cannot be subtracted; shares put both on a 0-1 scale. The difference is defined when one side is zero (common here), is bounded, sums to zero across categories, and reads directly in percentage points for a policy audience. The ratio $R_c = D_c/S_c$ is reported as a secondary index only, because it is undefined at $S_c = 0$ and explodes for tiny supply shares. A category with fewer than 3 postings or 30 graduates is labelled LOW SUPPORT and must not be presented as a finding.

4. Join, mapping and data treatment

Join key. Both sides are mapped to Week 4 taxonomy domain IDs (DOM-01 to DOM-15), which are stable concept IDs, and joined on `category_id`. Domains were chosen over occupations because supply records are programmes, which do not map to ISCO occupations, and because Week 5 found no demand-side skill text to join at skill level (posting descriptions are empty).

Mapping. Versioned regular-expression rules (config/week6/mapping_rules.csv) map each posting title and programme title to domains, citing the taxonomy alias each rule relies on. Only the highest-priority matches are kept, so a specific rule ("public administration" to DOM-07) beats a generic one ("administration" to DOM-06), which avoids the domain over-matching Week 5 reported. Week 5 extraction is used as an independent cross-check: where it produced a reviewed occupation or skill, its domain agrees with the rule mapping for 14 of 17 records. Running the gap with Week 5 mappings first is sensitivity S5.

Multi-category records. A record matching k categories at the top priority contributes 1/k to each (fractional counting), so every record counts exactly once. 1 posting and 2 programme rows are split, for example "Pharmaceutical and Medical Sales Representative" (half DOM-04, half DOM-06).

Uncertain mappings. A mapping is uncertain when its rule is marked uncertain, when the record is split, or when a combined graduate count spans categories. 5 of 13 postings and 10 of 44 programmes are uncertain (listed in Unmatched_Uncertain). The largest is Bachelor of Laws (329 graduates) mapped to DOM-07, because taxonomy v0.1 has no legal domain. Sensitivity S1 drops all uncertain records.

Duplicates and missing data. Duplicates are detected by title + employer (demand), institution + ceremony + programme (supply) and by raw_hash; none were found (0 + 0). Possible re-posts by intermediaries (BrighterMonday Consulting, Anonymous Employer) are kept but flagged and tested in S3. Records with no title or no graduate count would be excluded and listed (none here). Approximate counts (all IUIU rows) are kept and tested in S4. Unmatched: 0 postings and 0 programmes failed to map; categories with no evidence on either side: DOM-01, DOM-03, DOM-08, DOM-13, DOM-14, DOM-15.

5. Worked example: DOM-10 Transport, logistics and supply chains

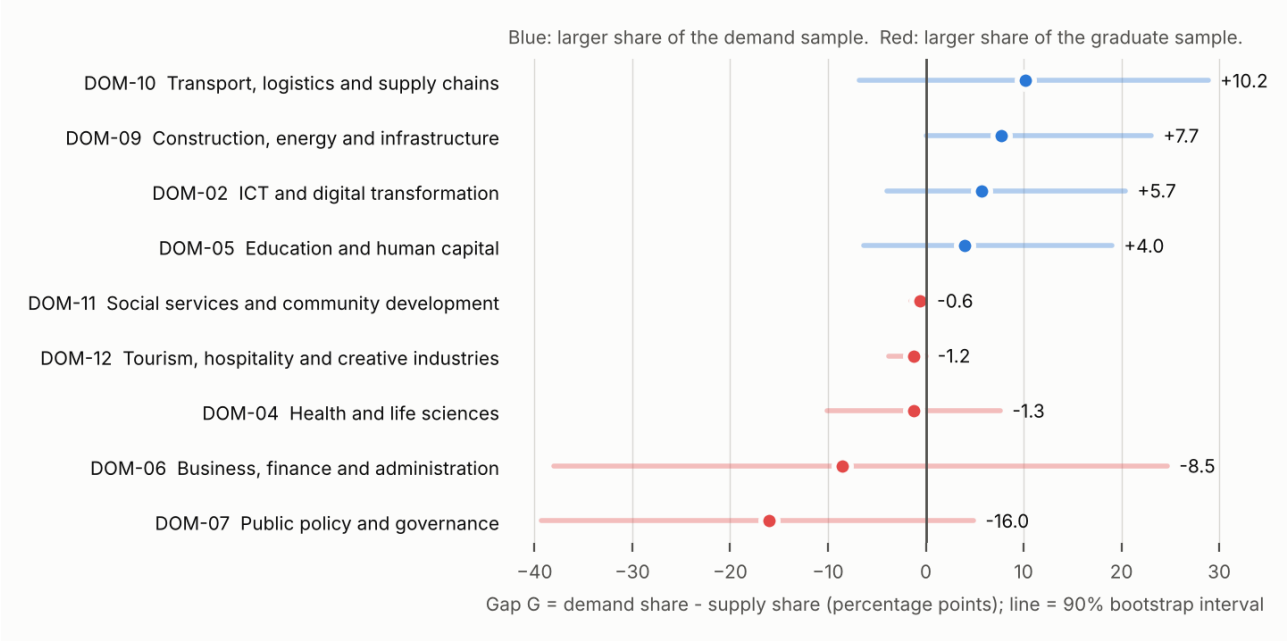
Step	Value	Source
Postings mapped to DOM-10	2	Procurement Officer; Stock/Inventory Manager (rules D01, D02)
All mapped in-scope postings	13	Coverage sheet
Demand share D	15.4%	2 / 13
Graduates mapped to DOM-10	187	UMI PGD Procurement and Supply Chain (127) + PGD Logistics and Transport (60), rule S08
All mapped graduates	3,585	Coverage sheet
Supply share S	5.2%	187 / 3,585
Gap G = 100(D - S)	+10.2 pp	Category_Results row DOM-10
90% bootstrap interval	-6.8 to +28.9	Crosses zero
Support	LOW SUPPORT	Only 2 postings

Each row traces to record IDs, source URLs and retrieval dates in the Joined_Sample sheet. The Week 4 taxonomy lists procurement under DOM-06 skills (SKL-038) but logistics and supply chains under DOM-10; the rule follows the DOM-10 alias, which is a documented taxonomy choice.

6. Results by category

Category	Posts	D	Grads	S	Gap pp	90% interval	Support
DOM-10 Transport, logistics and supply chains	2	15.4%	187	5.2%	+10.2	-6.8 to +28.9	LOW SUPPORT
DOM-09 Construction, energy and infrastructure	1	7.7%	0	0.0%	+7.7	+0.0 to +23.1	LOW SUPPORT
DOM-02 ICT and digital transformation	1	7.7%	71	2.0%	+5.7	-4.0 to +20.4	LOW SUPPORT
DOM-05 Education and human capital	1	7.7%	134	3.7%	+4.0	-6.3 to +19.1	LOW SUPPORT

Category	Posts	D	Grads	S	Gap pp	90% interval	Support
DOM-11 Social services and community development	0	0.0%	20	0.6%	-0.6	-1.5 to -0.1	LOW SUPPORT
DOM-12 Tourism, hospitality and creative industries	0	0.0%	43	1.2%	-1.2	-3.8 to +0.0	LOW SUPPORT
DOM-04 Health and life sciences	0.5	3.8%	183	5.1%	-1.3	-10.1 to +7.6	LOW SUPPORT
DOM-06 Business, finance and administration	6.5	50.0%	2,098	58.5%	-8.5	-38.1 to +24.7	OK
DOM-07 Public policy and governance	1	7.7%	849	23.7%	-16.0	-39.3 to +4.9	LOW SUPPORT
No evidence either side: DOM-01, DOM-03, DOM-08, DOM-13, DOM-14, DOM-15	0	0%	0	0%	0		NO EVIDENCE
Total (denominator)	13	100%	3,585	100%	0		



Only DOM-06 Business, finance and administration has adequate support; its gap is negative but its interval runs from -38.1 to +24.7 pp and its sign flips in 2 of 7 scenarios. The DOM-11 interval excludes zero only because it has 0 postings and 20 graduates; it is LOW SUPPORT and not a finding.

7. STEM-to-humanities comparison with NDPIV para 441(iv) (cited as 441(ii) in the brief)

Source text. NDPIV paragraph 441, item (iv) (printed p. 200, PDF p. 237) sets the programme result "Increased ratio of Science and Technology graduates to humanities graduates from 2:5 in FY2022/23 to 3:5". Item (ii) of the same paragraph is the unemployment target (11.9% to 8.1%), so the brief's "441(ii)" appears to refer to item (iv). The benchmark is therefore a ratio of *graduates*, with a FY2022/23 baseline. The Plan is not consistent elsewhere: printed p. 8 (PDF p. 45) describes a ratio of humanities to science *courses* of 2:5, and printed pp. 11 and 108 say the STEM-to-humanities graduate ratio has stagnated at 2:5. We use the programme result in 441(iv) as the operative definition.

Measure	STEM	Humanities	STEM per 5 humanities	Comparable?
Primary: supply, health excluded	71	3,331	0.11 : 5	NO - two-institution convenience sample
Health counted as STEM	254	3,331	0.38 : 5	NO - two-institution convenience sample
Health counted as humanities	71	3,514	0.10 : 5	NO - two-institution convenience sample

Measure	STEM	Humanities	STEM per 5 humanities	Comparable?
Exact counts only (drops approx/combined)	0	2,618	0.00 : 5	NO - two-institution convenience sample
Institution: Islamic University in Uganda	6	580	0.05 : 5	NO - two-institution convenience sample
Institution: Uganda Management Institute	65	2,751	0.12 : 5	NO - two-institution convenience sample
Demand analogue: postings in STEM vs non-STEM categories	2	10.5	0.95 : 5	NO - counts postings, not graduates
NDPIV baseline FY2022/23 (441(iv))			2 : 5	Policy benchmark
NDPIV target 2029/30 (441(iv))			3 : 5	Policy benchmark

Result. The supply sample has 71 STEM and 3,331 humanities graduates (health held out), a ratio of 0.11:5, roughly a fifth of NDPIV's 2:5 baseline. Counting the 183 health graduates as STEM raises it to 0.38:5, still below 2:5.

Why it diverges, and why it is labelled non-comparable. (1) Selection: UMI is a management institute and IUIU's Kampala campus mainly awards law, business and administration, so the sample excludes the universities where most science and engineering graduates qualify. (2) Level and period: most UMI awards are postgraduate diplomas from 2025 ceremonies, while the NDPIV baseline is a national FY2022/23 figure for all graduates. (3) Definition: the sample uses Week 3's provisional STEM/NON_STEM flag and treats all NON_STEM as humanities, which MAP-004 warns against; health is unresolved (MAP-006). (4) Counts: all IUIU counts are approximate, and both UMI ICT rows are combined counts. The comparison therefore tests the engine, not the policy claim. The demand analogue (0.95:5, counted in postings) is shown only to contrast the two mixes and is not an NDPIV measure. NDPIV also names "Science and Technology", not STEM; whether health and agriculture count as science is not defined in 441(iv), which is why the health variants are reported.

8. NDPIV ICT demand statement (cited as 9274 in the brief)

Source text. The statement sits in paragraph 383 (printed p. 177, PDF p. 214), on human resource capacity for the digital economy. It says ICT graduates are not yet meeting private-sector demand and lack practical skills in AI, blockchain, cloud computing and cybersecurity. The same paragraph names UMI among the institutions training ICT professionals, and notes Makerere alone produced over 2,100 ICT graduates in five years. NDPIV has no paragraph 9274; the number most likely refers to a line in a text copy of the Plan.

In this sample 1 of 13 postings is an ICT role (MIS Associate) and 71 of 3,585 graduates hold ICT awards, a gap of +5.7 pp with interval -4.0 to +20.4 pp. One posting cannot support any statement about ICT demand. Paragraph 383 is also a claim about skills quality, not numbers: testing it needs employer-required skill mentions such as cloud computing (SKL-013), machine learning (SKL-007) and cybersecurity (SKL-020), all already in the Week 4 taxonomy, extracted from posting descriptions. Week 5 found every description empty, so the data cannot test the statement until posting bodies are collected.

9. Robustness checks

Bootstrap. Postings and programme rows were resampled independently with replacement (2000 times, seed 20261106) to give 90% intervals. Their width (tens of percentage points) is the main reasonableness check: with 13 postings, one posting moves a demand share by 7.7 pp.

Category	S0	S1	S2	S3	S4	S5	S6	Sign holds
DOM-07	-16.0	-15.3	-10.3	-12.6	-8.6	-16.1	-26.9	7 of 7
DOM-06	-8.5	-16.8	-8.5	-19.6	-23.7	+1.8	+10.7	5 of 7
DOM-04	-1.3	-5.8	-1.8	+0.5	+3.8	-5.1	-3.6	5 of 7
DOM-12	-1.2	+0.0	-1.2	-1.2	+0.0	-1.2	-1.8	5 of 7
DOM-11	-0.6	-0.4	-0.6	-0.6	-0.3	-0.4	-0.8	7 of 7
DOM-05	+4.0	+9.0	+2.9	+7.4	+5.1	+5.2	+2.2	7 of 7
DOM-02	+5.7	+10.2	+4.7	+9.1	+7.7	+5.7	+4.8	7 of 7
DOM-09	+7.7	+0.0	+6.7	+11.1	+7.7	+0.0	+7.7	5 of 7

Category	S0	S1	S2	S3	S4	S5	S6	Sign holds
DOM-10	+10.2	+19.0	+8.1	+5.9	+8.2	+2.5	+7.8	7 of 7

S0 primary; S1 certain mappings only; S2 include jobs outside Uganda; S3 exclude intermediary employers; S4 exact supply counts only; S5 Week 5 extraction mapping first; S6 drop the two largest programmes (PGD Project Planning 578, PGD M&E 556). DOM-06's sign depends on the choices (S5, S6), so its direction is not robust. DOM-07 stays negative in every scenario, driven by public administration and law graduates, but its interval still crosses zero.

10. Limitations and bias

• **Convenience sample, not a national estimate.** 13 postings from one job board and one district, and graduates of two institutions. No figure here describes the Ugandan labour market or graduate output.

• **Source bias.** Online postings over-represent formal, urban, white-collar jobs; 4 of 12 BrighterMonday postings come from intermediaries. Supply is postgraduate- and management-heavy.

• **Informal-sector blind spot.** No posting evidences DOM-13 informal work, where most Ugandans work; the gap for informal skills is unknown, not zero.

• **Publication lag.** Supply is 2025 ceremonies, demand is October 2026; graduation lists appear months after ceremonies and national statistics (NCHE, UBOS) lag further.

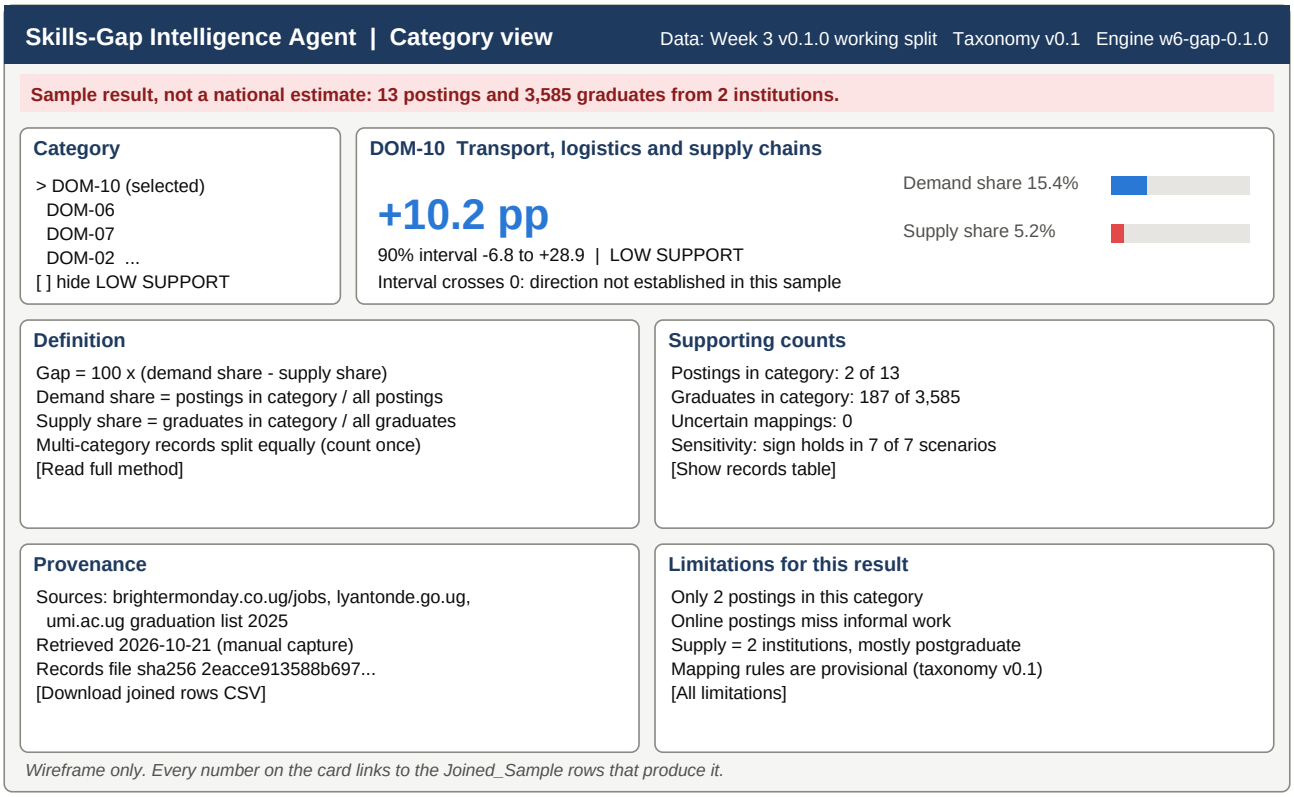
• **Extraction errors.** Week 5 v1 recall is limited to taxonomy aliases and posting descriptions are empty, so demand is measured by job titles only.

• **Taxonomy uncertainty.** Taxonomy v0.1 is provisional: no legal or media domain, skill parent IDs inconsistent with domain aliases (public administration), health unresolved. Mappings may change with v0.2.

11. Policy statements: allowed and to avoid

Allowed (cautious, traceable)	Must avoid in Week 7
"In the Week 3 sample, business and administration is about half of both postings and graduates."	"Uganda has a shortage of supply-chain graduates." (2 postings, interval crosses 0)
"The two sampled institutions produce far fewer STEM than humanities graduates (0.11:5)."	"Uganda's STEM ratio has fallen below the NDPIV baseline."
"The sample cannot test the NDPIV ICT-demand statement (para 383); posting descriptions are needed."	Any ranking of categories without its denominator, interval and support label

12. Week 7 interface wireframe



The card shows the result with its interval and support label, the definition in plain words, the counts behind both shares, provenance with a link to the record rows, and the limitations that apply to this category. LOW SUPPORT categories are hidden by default and the not-a-national-estimate banner cannot be dismissed.

13. Clean-run instructions

```
git checkout feature/week6-gap-engine

pip install pandas numpy pyyaml openpyxl matplotlib reportlab pytest

python src/skillsgap/gap/engine.py # outputs in data/processed/week6/

python src/skillsgap/gap/export_xlsx.py # reports/week6/TeamName_Week6_Analysis.xlsx

python src/skillsgap/gap/report_pdf.py # this report

python -m pytest tests/test_gap_engine.py # 5 tests, includes a reproducibility check
```

The engine reads the taxonomy from data/raw/taxonomy/ if unpacked, otherwise directly from the Week 4 zip in the repository root. To re-run on new Week 3 data, point inputs.records in config/week6/gap_engine.yaml at the new file; unmapped titles appear in unmatched_uncertain_excluded.csv for new rules.